ANDROID HANDHELD LAUNCHER · FEATURE OVERVIEW

Pavo.

Everything on the device,
under the thumbs.

VERSION 1.0.33

RGB LIGHTING EDITION

Package	Platform	Reference device	Licence
com.ben.pavo	Android 12 or newer	AYN Odin 3 · Android 15	Apache 2.0

November 2026 · github.com/Ben0019/pavo-launcher

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Feature areas

A complete replacement for the stock launcher, written against the Android platform with no AndroidX, no Gradle runtime and no root-only hooks.

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NEW IN 1.0.33

RGB, per stick

- **BOTH / EACH** switch: share one colour and mode, or give each stick its own.
- **Independent modes**: one stick can breathe while the other stays static.
- **Editable hex code** with type, delete, copy, paste and cut.
- **Invalid code** shows a toast and restores the last valid colour.
- **Screen mode** follows the left and right halves of the display separately.
- **Gesture toggles**: tap in BOTH, long-press in EACH.

Also in this build

Performance overlay text now stays fully opaque when the panel transparency is reduced; only the background fades. Charge separation from 1.0.32 keeps the configured limit instead of oscillating and overcharging.

Two rings. Two zones. Full control.

The Odin 3 exposes one SN3112 LED driver per joystick. Each stick has a single live RGB zone, so the launcher can paint two independent colours at once — one per hand — and every mode, colour and gesture is available from the RGB app.

What the hardware can do

- **One addressable zone per stick.** The second logical zone (handle ring) is not wired on the Odin 3.
- **Eight driver channels** all address the same physical zone; a single ring cannot show a gradient or two colours at the same time.
- **Two sticks, two colours** simultaneously — left and right are controlled independently.
- **Any mode on either stick**, including one animated and one static.

Controls

- **Brightness** 1–100% per frame, applied to both zones.
- **Speed** 10–300% for breathing, rainbow, strobe and multicolour.
- **Hue & saturation** sliders plus a full colour wheel.
- **Editable hex** code for exact colours.
- **PSERVER / DIRECT** transport badge; settings persist across restarts and screen wake.

MODE	BEHAVIOUR
STATIC	Fixed colour. Each stick keeps its own stored colour.
BREATH	Slow brightness pulse. Speed controls the cycle; colour is preserved.
RAINBOW	Continuous hue rotation across the spectrum.
STROBE	On/off flashes at the speed setting.
MULTI	Two hues at once, one per stick, both cycling with a 120° offset.
SCREEN	Follows the display. Left half drives the left stick, right half the right stick. Detailed on the next pages.
OFF	Disables that stick without forgetting its last colour or mode.

> **The honest limit.** A ring is one RGB zone. A screen with many colours can only be reduced to two simultaneous colours across the pair of sticks; the software selects the most representative colours rather than inventing a gradient the hardware cannot draw.

Per-stick editing

Tap a joystick to choose which stick the colour wheel, sliders and mode buttons edit. The selected stick is marked with an accent pill; BOTH keeps the pair together, EACH lets them diverge.

BOTH

- One colour and mode shared by both sticks.
- Switching to BOTH adopts the currently selected stick's settings for both.
- Tap either joystick to turn both off; tap again to restore their last mode.

EACH

- Independent colour and mode per stick.
- Tap selects the stick for editing.
- Long-press toggles just that stick off/on and remembers its last non-off mode.
- Linked mode syncs animation phases; EACH lets them run independently.

Hex code

Tap the code under the wheel to edit. Supports `#RRGGBB`, `RRGGBB` and `#RGB`.

- Type, delete, select, copy, paste and cut with the normal keyboard and selection handles.
- Done or tapping away commits; the wheel, sliders, previews and LEDs update together.
- Invalid code shows a toast and restores the last valid colour.



EACH mode: left STATIC red, right STATIC green. The left stick is being edited; the hex field is live.

Controller

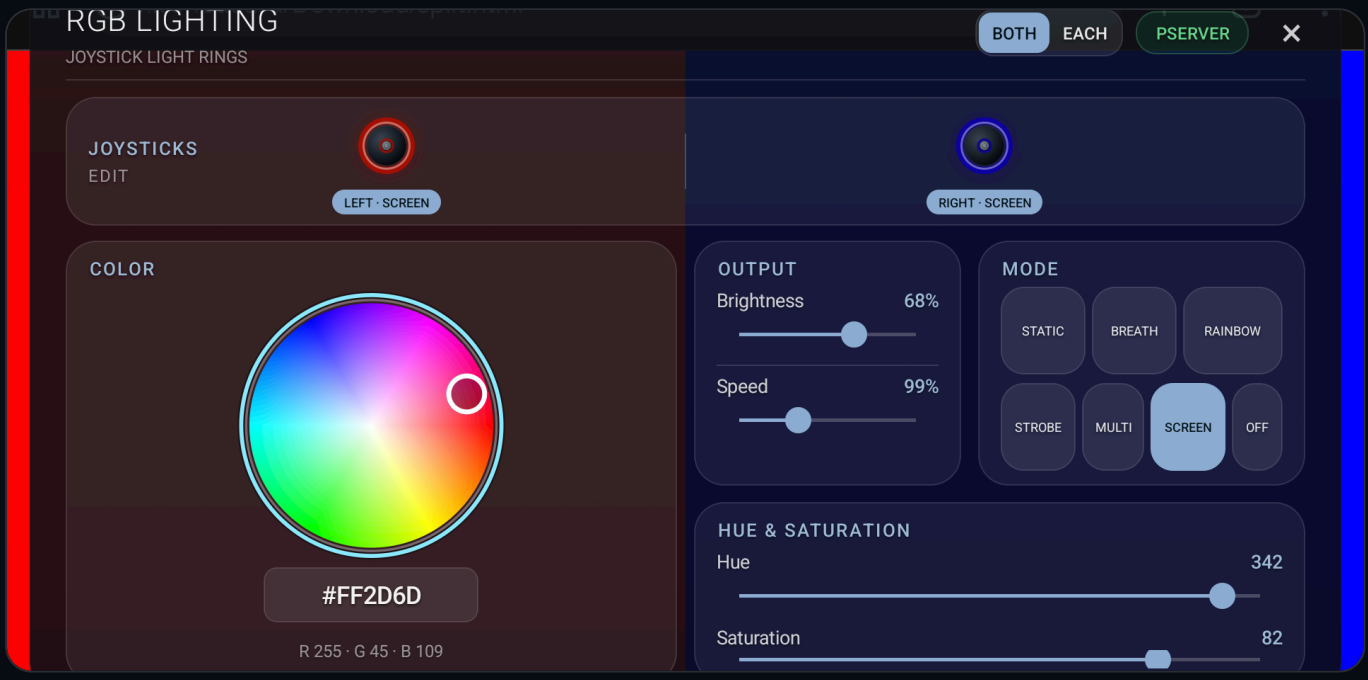
- D-pad / left stick moves focus; A selects the focused joystick.
- Hold A toggles the focused stick off/on.
- Focus rings, the accent selection pill and the mode highlight are all visible from the sofa.

SCREEN mode

SCREEN reads the current display through the privileged PServer channel and paints each stick with its side of the screen. It is designed for ambient, moment-to-moment matching rather than a colour show.

How it works

- **Left half → left stick; right half → right stick.**
- **Dominant hue** per half, blended with 22% of that half's average colour so it still reads like the region.
- **Grey and white stay grey/white**; very dark or letterboxed regions go dark.
- **160 ms sampling** and a 0.38 per-frame blend: a scene change settles in roughly 0.25 s without a hard snap.
- **Pauses while the RGB app is open** so it never samples its own UI.
- **Works in BOTH and EACH.** If only one stick is in SCREEN, it follows its own side while the other keeps its mode.



SCREEN mode over a split red/blue display: the left ring shows red, the right ring shows blue. The hex field and mode buttons remain available for the other modes.

Accuracy choices

- Dominant hue voting ignores large neutral areas, so a vivid accent is not averaged into grey.
- 22% of the plain average is mixed back to keep the actual tint.
- No palette cycling: the result is stable while the screen is stable.

What it cannot do

- Show a spatial gradient around one ring; the ring is a single zone.
- Show more than two simultaneous colours across both sticks.
- Capture DRM-protected or secure surfaces.

THE REST OF PAVO

A complete handheld home

01 · Home screen

- Paged icon grid, folders, resizable widgets.
- Clock, weather, status header and transparent dock.
- Layout backup and restore as a setup ZIP.

02 · App drawer & navigation

- Full drawer and dock pop-up drawer.
- Predictive back, install shortcuts, focus-first D-pad navigation.

03 · Appearance & wallpaper

- Liquid Glass dock, drawers and built-in sheets.
- Wallpaper-derived accent, icon packs, per-surface transparency.
- OLED protection with pixel shift and idle dim.

04 · Performance tuning

- Per-cluster clocks, governors and GPU frequency table.
- Presets and per-app profiles; background hold; thermal guard.

05 · Performance overlay & logs

- CPU/GPU temp, clock, load, RAM, FPS graph, per-core rows.
- Battery temperature and power draw.
- Opacity affects the background only; text stays fully opaque.

06 · Stress testing

- CPU, GPU and combined runs; fixed clocks or tuning profile.
- Live graph, averages, maxima, throttle and hold percentages.

07 · Fan control

- Off, Low, Smart, High, Curve and Manual.
- 15-point curve, CPU/GPU/BOTH source, per-app profiles.

08 · Device health & charging

- Battery model, health, cycles, resistance, USB details.
- 80% limit and automatic charge separation.

09 · File manager

- Dual panes for internal, SD and USB OTG.
- Copy, cut, paste, rename, delete, ZIP, previews and text editing.

10 · FTP host & client

- Share storage with read/write switches and QR login.
- Mount a remote server as a pane; live previews and transfers.

12 · Controllers & shortcuts

- System-wide remapping, per-app layouts, touch mapping.
- In-game overlay menu and physical shortcuts.

13 · Setup & privacy

- Guided first-run permissions and home role.
- PServer only; read-only fallback; local by design.

14 · Languages & limits

- 25 locales; Android 12+; 1920×1080 and 1280×960, portrait fallback.
- Clock and fan writes remain requests to the firmware.

11 · RGB lighting

- Two independent zones, seven modes, per-stick editing.
- Editable hex, gesture toggles, screen-follow colour.

CONTROLLER & INSTALL

The ring never leaves you stranded

Home & drawers

- **A** — open the focused app, or activate the focused control.
- **Hold A** — item menu: rename, icon, info, remove, folder.
- **B** — back out of a menu, folder or drawer.
- **L1 / R1** — previous / next page; also page edges in lists.
- **L2** — dock pop-up drawer; **R2** — full drawer.

RGB & system shortcuts

- **A** on a joystick selects it; **Hold A** toggles it off/on.
- **BTN_C** — floating side drawer; hold for its style menu.
- **BTN_Z** — performance overlay; hold for the overlay menu.
- **Perf Logs key** — optional third shortcut.

GETTING STARTED

01
Install `Pavo.apk` or run `adb install -r Pavo.apk`.

02
Open Pavo once and grant or skip each permission in the setup sheet.

03
Set Pavo as the home app, then press Home.

Compatibility & limits

- Android 12+ (API 31), 25 languages, 1920×1080 and 1280×960 with portrait fallback.
- Privileged writes use the vendor PServer channel; no su, Magisk or KernelSU dependency.
- One RGB zone per stick: two simultaneous colours maximum.
- Clock and fan writes can still be reduced by the vendor thermal manager.

2	7	25
RGB ZONES	LIGHTING MODES	LANGUAGES

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